

ZOMBIE: STRING OF SURVIVAL

THE DEFINITIVE COMPLETE MASTER RE-CREATION PROMPT & SPECIFICATION

Target Engine: Three.js (r128) WebGL Web Game | **Prompt Mode:** 100% Zero-Loss Single Paste Re-Creation

CORE NARRATIVE PREMISE:

"You are Alexia, trapped in a zombie-infested District 4, and you have 7 days to explore the city, collect resources, rescue survivors, manage your increasingly detectable base, adapt to enemies that respond to your playstyle, uncover the infection mystery, and finally escape from the harbor."

SECTION 1: THE ONE-SHOT MASTER PROMPT (COPY-PASTE READY)

The boxes below form the complete, unified master prompt. You can copy and paste all parts together into any coding AI to re-create the exact game with 100% fidelity:

PROMPT PART 1: CORE ARCHITECTURE, PROTAGONIST RIG (ALEXIA) & ATMOSPHERIC TREES

Build a complete, production-ready AAA 3D/4D/5D Survival Horror Web Game titled 'ZOMBIE: STRING OF SURVIVAL'.

PREMISE: You are Alexia, trapped in a zombie-infested District 4, and you have 7 days to explore the city, collect resources, rescue survivors, manage your increasingly detectable base, adapt to enemies that respond to your playstyle, uncover the infection mystery, and finally escape from the harbor.

TECH STACK: Vanilla HTML5, CSS3, JavaScript with Three.js (r128) bundled locally in 'js/three.min.js'. All 3D characters, survivors, and zombies MUST be constructed using procedural Three.js composite meshes and materials so the game requires zero external 3D model downloads (no .gltf/.fbx), loads instantly offline/Vercel, and avoids CORS issues.

ALEXIA RIG: Build playerGroup at (-25, 0, -10). Protagonist is Alexia, an athletic female survivor. Torso: ivory white sleeveless blouse/tunic (BoxGeometry 0.52, 0.52, 0.32, color 0xfcfbfa) with skin décolletage opening (0.20, 0.22, 0.04 color 0xf5c2a3) and draped V-neck lapel folds angled 0.25 rad. Cinched natural waist with white fabric sash (0.48, 0.11, 0.30), circular silver ring buckle (TorusGeometry 0.045, 0.012), cascading tie ends (0.07, 0.22, 0.03), and draped peplum tunic hem (CylinderGeometry 0.24, 0.29, 0.24) cascading over hips. Cross-body slim utility sling pack (0.32, 0.26, 0.12 color 0x3f3f46) with diagonal chest strap, plus compact VHF emergency radio with 0.22m whip antenna on side belt. Head & Hair: delicate feminine head (SphereGeometry 0.20 scaled 0.88, 1.05, 0.94 color 0xf5c2a3), soft chin contour, almond brown eyes (iris 0x4a2c11) with white sclera and dark upper eyelashes, arched brows, small nose, and soft rosy lips (0xc87070). SIGNATURE HAIR: Long flowing chestnut auburn wavy hair (0x5c3418) parted cleanly in the center, with crown dome, cascading wavy locks falling over BOTH shoulders to mid-chest (dual cylinder curls 0.36m & 0.32m each side), and long cascading locks down the back (0.40m & 0.42m). Limbs: bare feminine arms (upper arm 0.062, forearm 0.052), silver wrist bracelet tracker, TPS weapon rig. Fitted dark indigo denim skinny jeans (0x182435, thighs 0.10, calves 0.08) with articulated knees. Cream espadrille wedge flats (0xf3ede2) with woven jute sole trim (0xd2b58d) and delicate ankle straps with buckles. Camera: auto-hide head in FPS mode (V) to avoid camera clipping; reveal complete character in TPS (V) and Drone (T) views with weapon pitch aiming.

ATMOSPHERIC DECIDUOUS TREES: Construct procedural broadleaf deciduous trees matching reference plant photography across District 4. Base trunk rises to 4.0m with grooved vertical fissured bark texture (CanvasTexture) and 4 root flares extending into a dark soil mound. At 4.0m, the trunk bifurcates into dual ascending boughs (left bough angled 0.18 rad, right bough angled -0.22 rad) that fork into 7 secondary branches, including a prominent weeping lower-right bough. The canopy forms an expansive rounded umbrella spread (8-10m width) consisting of 16 compound foliage clusters in 3 tones of forest green (0x1b2e19, 0x243e21, 0x31522a) with double-sided alpha leaf branch sprite planes on outer tips. In the render loop, animate gentle canopy wind sway using sinusoidal phase offsets.

ARCHITECTURAL 3-STORY BRICK TOWNHOUSES: Construct procedural urban brick buildings matching reference architectural photography across District 4 and as the Safehouse Headquarters. Each structure features: (1) Chamfered concrete foundation plinth (0x64748b, height 1.2m) at ground level; (2) Warm terracotta red brick masonry main body (0xb55335) with running bond mortar texture (CanvasTexture); (3) Protruding ground-floor stone storefront bay (0x857d77) with 4 vertical ashlar pilasters, 2 large multi-pane shop display windows with wooden cross mullions, central white paneled entrance door with brass knob, and projecting molded stone entablature cornice slab; (4) Front upper facade with 6 arched windows (3 per floor) featuring white multi-pane frames, stone arched headers/lintels, and stone sills; (5) Side elevation with rows of rectangular sash windows, arched wooden door with stone arch surround, 4-step concrete stoop stairway leading down to pavement, and vertical metal drainage downspout pipe; (6) Flat roof with gravel/cracked asphalt deck (0x4b5563), raised perimeter parapet walls with weathered stone coping on all 4 edges, red brick chimney stack with stone capping rim, and rooftop industrial HVAC condenser compressor unit with twin curved duct elbows.

PROMPT PART 2: NPC SURVIVORS & HORROR ZOMBIE ANATOMY**NPC SURVIVORS (3 Unique Characters):**

1. Dr. Evelyn Reed (Doctor, 50, -48): White lab coat (0.64, 0.92, 0.40) over turquoise scrubs (0x0d9488), stethoscope torus tube with silver chestpiece disc, Red Cross armband (0xef4444), wireframe glasses, ponytail hair bun, light blue surgical gloves, holding illuminated cyan antiviral sample vial.
2. Marcus Vance (Engineer, -34, -42): Safety orange boiler suit (0xea580c) with grease smudges, dual brass welding goggles on forehead, thick dark brown beard, heavy leather toolbelt with 3D pipe wrench (0.42m cylinder + box head) and hammer, steel-toed work boots.
3. Sgt. Darius Cole (Soldier, -22, -22): Digital woodland camo BDU (0x3f4f2e) with tactical assault vest (0x1f2937), Kevlar helmet with NVG mount and chinstrap, facial combat scar over eye, slung assault rifle (0.10, 0.16, 0.85), military jump boots.

Survivors feature idle breathing, head tracking that turns to look at Alexia within 8m, and overhead rotating role beacons.

HORROR ZOMBIES (Visceral Undead Anatomy & 4 Archetypes):

Necrotic Flesh & Posture: Sickly olive-green skin (0x44533c, roughness 0.92). Torso hunched forward 0.38 rad (25 deg) with 5 protruding spinal vertebrae bone spheres (0.055) along back. Gaping chest wound (0.36, 0.45, 0.08, color 0x5e0b0b) with 3 curved TorusGeometry rib bones protruding out with dripping blood.

Gaunt Skull & Snarling Jaw: Emaciated skull with hollow black sockets, cranial trauma with exposed white bone, and glowing infected eyes. Separate lower jaw hinged open 0.35 rad in a snarl with dark throat interior and dual rows of 7 sharp yellow bloody cone teeth (0.018, 0.06).

Claws & Bare Foot: Reaching arms rotated forward -1.15 rad with claw hands having 4 splayed bony necrotic fingers + thumb with cone talons. Left leg has exposed white kneecap bone sphere in torn knee; right leg ends in bare rotting foot with 5 blackened toe boxes dragging on pavement.

4 Archetypes: Hunter (standard walker, speed 3.2, HP 55), Swarm (feral sprinter, speed 5.2, HP 42, 4 mutated spine bone spikes, rapid sprint), Ambusher (speed 4.6, HP 38, dark chitin back plates, elbow bone spurs, glowing cyan eyes), Tank (1.85x scale, HP 240, 6 pulsating toxic boils 0x84cc16, embedded rebar, colossal sledge-fist).

Animations & FX: Asymmetric shambling limp (one leg strides while bare foot drags), head twitches, vicious clawing slash attacks when close (<2.8m), mesh damage flash (0xff2222), and ragdoll collapse on death with an expanding dark blood pool decal (CircleGeometry radius 1.15-2.4m, color 0x3d0707).

PROMPT PART 3: COMBAT, BASE HEAT, WEB AUDIO SYNTH & MULTI-TAB INTERFACE

COMBAT & WEAPONS:

WASD move, Shift sprint (stamina drain, 70 noise), C crouch (cuts speed to 3.5, cuts noise to 15, reduces detection radius), Mouse look (Pointer Lock + drag fallback), Left Click attack, Right Click ADS zoom (FOV 65 down to 42), 1 for Pistol, 2 for Barbed Bat, R to reload, E to interact, T for Drone, V for 1st/3rd View.

Suppressed Pistol: 24 rounds, slide recoil, ray-cone hitscan detection (dot>0.85 & lateralDist<2.4m or dist<4.5m & dot>0.45), hit marker overlay (X), muzzle flash.

Barbed Bat: 120 melee damage, wide swing arc animation, barbed wire coils.

BASE HEAT & INFECTION DOME:

Safehouse generator at (-40, 0, -40) generates an expanding pulsating orange wireframe heat dome (radius = heat * 2.0, up to 90m+) that attracts distant zombies to base.

Zombie hits increase infection (0-100%), dynamically driving sensory degradation: low-pass filter sweeps down from 22kHz to 500Hz, heartbeat speeds up, red bio-veins creep into view.

WEB AUDIO API SYNTHESIZER (audio.js):

Zero external audio files: 58Hz sawtooth generator rumble, heartbeat pulses (65Hz->35Hz sine ramps), gunshot noise bursts (exponential 1200Hz->80Hz lowpass filter), zombie growls (sawtooth sweep), fleshy impacts, and master lowpass filter cutoff.

HUD & MULTI-TAB INTERFACE:

Tab 01: 3D Survival World with top compass bar (0 deg | N), action toolbar (Drone, 1st/3rd View, Pistol, Bat, Restart, Exit Game [ESC]), crosshair, Hunted warning banner, damage flash vignette, game over modal, and 'You are out!' Exit Popup.

Tab 02: DAG Decision Tree interactive canvas graph tracking committed nodes and branches.

Tab 03: Adaptive AI Director Radar visualizing Stealth, Combat, and Velocity metrics.

Tab 04: Thermal Heat Dissipation 2D Radar Canvas.

Tab 05: Sensory Infection Bio-Vein Distortion Canvas with FOV indicator.

Tab 06: Tactical 2D Top-Down Arena Simulator with flashlight cone.

SESSION EXIT & LIFECYCLE MANAGEMENT (onExit, Reset & MainMenu):

1. Detect Exit Event: Listen for 'onExit' custom event, window blur, visibilitychange (leaving screen), pointerlock release, and ESC key / Exit HUD button.
2. Show Exit Popup: Display 'You are out!' modal overlay with frosted glass blur over the frozen screen, current run statistics, and a 3-second auto-timer countdown with a manual continue button.
3. Reset Game State: resetGameState() purges all corpses, resets score (0 kills), level (Day 1), vitals (100 HP, 100 Stamina, 0% Infection, 24 Ammo), stops timers and audio hums, and frees WebGL resources.
4. Return to Main Screen: loadMainMenu() closes the popup, unblocks all buttons, and returns to the opening screen.
5. Start New Game: initGame() launches a fresh session seamlessly with pointer lock and audio initialized.

Server: Node.js server.js serving static files on port 3000, and vercel.json for direct deployment.

SECTION 2: COMPONENT SPECIFICATIONS & MATHEMATICAL BLUEPRINTS

This reference section documents exact coordinate offsets, geometry dimensions, and algorithms:

2.1 Alexia Protagonist Model Blueprint

- **Head & Facial Structure:** SphereGeometry(0.20, 18, 16) scaled (0.88, 1.05, 0.94) color 0xf5c2a3 (warm feminine skin). Feminine chin sphere (0.08) at z=0.13. Dual almond eyes with white sclera, warm brown irises (0x4a2c11), dark pupils, upper eyelash bars (0x1c1917), and gracefully arched brows (0x452311). Delicate nose cone and soft rosy lips (0xc87070) with curved upper and lower contours.

- **Long Flowing Chestnut Auburn Hair:** Signature hair color 0x5c3418. Crown dome SphereGeometry(0.225) with center scalp part. Cascading locks down back (dual cylinders 0.40m & 0.42m to mid-back). Cascading wavy locks falling over BOTH left and right shoulders to mid-chest (dual angled cylinders 0.36m & 0.32m each side).
- **Torso & Flowing Blouse:** Ivory white sleeveless tunic/blouse (BoxGeometry 0.52, 0.52, 0.32 color 0xfcfbfa) with skin décolletage opening (0.20, 0.22, 0.04) and draped V-neck lapel folds angled 0.25 rad. Cinched natural waist with fabric sash (0.48, 0.11, 0.30), circular silver ring buckle (TorusGeometry 0.045, 0.012), cascading tie ends (0.07, 0.22, 0.03), and draped peplum hem (CylinderGeometry 0.24, 0.29, 0.24) cascading over hips.
- **Articulated Limbs & Denim:** Bare feminine arms (upper arm 0.062, forearm 0.052), silver wrist bracelet / tracker, TPS weapon rig. Fitted dark indigo denim skinny jeans (color 0x182435, thighs 0.10, calves 0.08) with articulated knees. Cream espadrille wedge flats (0xf3ede2) with woven jute sole trim (0xd2b58d) and delicate ankle straps.

2.2 Horror Zombie Anatomy Blueprint

- **Necrotic Anatomy:** Sickly olive-green skin 0x44533c (roughness 0.92). Torso hunched forward 0.38 rad (25 deg) with 5 protruding spinal vertebrae spheres (0.055) along back. Gaping chest wound (0.36, 0.45, 0.08) color 0x5e0b0b with 3 curved TorusGeometry(0.14, 0.022, 6, 8, 0.65*PI) rib bones protruding outward.
- **Gaunt Skull & Snarling Jaw:** Hollow black eye sockets with glowing infected eyes (Hunter: red 0xef4444, Swarm: amber 0xf59e0b, Ambusher: cyan 0x00f5d4, Tank: green 0x22c55e). Separate lower jaw hinged at z=0.05, rotated open 0.35 rad with bloody throat cavity and dual rows of 7 sharp cone teeth (0.018, 0.06) color 0xfef08a.
- **Claws & Bare Foot:** Reaching arms rotated forward -1.15 rad with claw hands having 4 splayed bony fingers + thumb with cone talons. Left leg has exposed white kneecap bone sphere (0.065) in torn knee; right leg ends in bare rotting foot with 5 blackened toe boxes dragging on pavement.
- **Mutant Archetypes:** Swarm has 4 mutated cone bone spikes (0.35m) along spine; Ambusher has dark chitin back plates and elbow bone spurs; Tank is scaled 1.85x, has 6 glowing toxic boils (0x84cc16), embedded rebar, and colossal sledge-fist BoxGeometry(0.34, 0.38, 0.32).

2.3 Atmospheric Deciduous Tree Blueprint

- **Trunk & Bifurcated Boughs:** Base trunk CylinderGeometry(0.38, 0.52, 4.0) rising from 4 root flare cones ConeGeometry(0.18, 0.95) and soil mound. At 4.0m height, bifurcates into dual boughs: Left Bough CylinderGeometry(0.24, 0.34, 4.4) angled 0.18 rad, and Right Bough CylinderGeometry(0.26, 0.36, 4.8) angled -0.22 rad. 7 secondary structural branches fan outward, including a prominent weeping lower-right bough angled -0.78 rad.
- **Grooved Bark Texture:** Procedural CanvasTexture (256x512) with warm umber gradient, 55 vertical rough fissures and shadow crevices, 35 lighter bark ridges, and micro-noise striations.
- **Canopy & Foliage Clusters:** 16 compound DodecahedronGeometry(r, 1) foliage clusters in 3 tones of forest green (0x1b2e19, 0x243e21, 0x31522a) scaled into an umbrella canopy with drooping right flank, supplemented by 8 cross-quad leaf branch sprite planes using procedural pinnate leaf textures with alpha cutout.
- **Wind Sway:** Canopy groups animated in render loop with sinusoidal phase modulation for gentle natural breeze sway.

2.4 Architectural 3-Story Brick Townhouse Blueprint

- **Foundation Plinth & Masonry:** Chamfered concrete foundation plinth BoxGeometry(w + 0.6, 1.2, d + 0.6) color 0x64748b at ground level with top stone drip-lip edge. Warm terracotta red brick masonry main body BoxGeometry(w, h, d) color 0xb55335 with procedural running-bond canvas texture (mortar seams, color variation, weathered brick noise).
- **Ground-Floor Stone Storefront Bay:** Protruding ashlar stone storefront bay BoxGeometry(w * 0.94, h * 0.30, 2.0) color 0x857d77 with stone masonry texture, 4 vertical rectangular stone pilasters, 2 large multi-pane shop display windows with wooden mullion cross-grids (color 0x93c5fd/0x475569), central white paneled entrance door with brass knob, and a projecting molded stone entablature cornice slab.
- **Front Arched Windows:** Floors 2 and 3 each feature 3 arched windows (6 total) with multi-pane white window frames, curved semicircular stone arch headers (TorusGeometry), and stone sill ledges (BoxGeometry).
- **Side Elevation (Door, Stoop, Windows, Downspout):** Rows of rectangular sash windows with stone sills, an arched wooden side entrance door (0x78350f) with semicircular stone arch surround, a 4-step concrete stoop stairway leading down to the pavement, and a vertical metal drainage downspout pipe (0x475569) with upper roof hopper.
- **Roof Architecture (Parapets, Chimney & HVAC):** Recessed flat roof deck with cracked weathered gravel/asphalt texture (0x4b5563). 4 raised perimeter parapet walls with molded stone coping slabs on all edges. Red brick chimney stack (BoxGeometry) with stone capping rim. Industrial rooftop HVAC condenser compressor unit (BoxGeometry 0xd1d5db) with twin curved duct elbows.

(TorusGeometry 0x9ca3af) and cooling fan vent.

SECTION 3: COMPETITION BREAKDOWN & FEATURE MATRIX

For competition submissions and technical reviews, this matrix defines what features are fully playable in the real-time 3D engine versus telemetry and conceptual references:

System Component	Runtime Status	Playable In-Game Functionality
3D Survival Horror Engine	100% Implemented	Three.js WebGL city district, day/night cycle, burning embers, lighting
Alexia Player 3D Rig	100% Implemented	White belted tunic, ring buckle, long chestnut hair, skinny jeans, espadrilles, TPS weapons
Deciduous City Trees	100% Implemented	Bifurcated trunk, grooved bark, 16 foliage clusters, drooping branch, wind sway
3-Story Brick Townhouses	100% Implemented	Concrete plinth, stone storefront, arched windows, side stoop, parapet, chimney, HVAC
Exit Detection & Main Menu	100% Implemented	onExit event, 'You are out!' popup, frozen screen, state reset, loadMainMenu(), initGame()
NPC Survivors (3 Roles)	100% Implemented	Dr. Reed, Marcus, Sgt. Cole models with idle breathing, head tracking, quests
Horror Zombies (4 Types)	100% Implemented	Rotting flesh, exposed ribs, snarling teeth, bare foot, claw attacks, blood decals
FPS / TPS / Drone Camera	100% Implemented	1st person, 3rd person with weapon pitch aim, 80m reconnaissance drone
Combat & Hitscan Detection	100% Implemented	Suppressed pistol, barbed bat, recoil, hit markers, damage flashes
Base Heat Attraction Dome	100% Implemented	Pulsating orange wireframe dome attracting zombies into safehouse
Web Audio Synthesizer	100% Implemented	Generator rumble, heartbeat, gunshots, zombie growls, infection muffling
DAG Narrative Decision Tree	UI Simulation	2D interactive canvas graph tracking commitment paths and branch nodes
AI Director Radar	UI Telemetry	Tri-axis player playstyle telemetry (Stealth, Combat, Velocity)
Thermal Heat Dissipation Radar	UI Radar	2D top-down heatmap simulation of acoustic and thermal dissipation
Sensory Vein Distortion	UI Post-Process	Procedural red bio-vein canvas simulating peripheral tunnel vision
Tactical 2D Arena	Playable 2D Fallback	HTML5 top-down 2D canvas survival simulator with flashlight cone

SECTION 4: 13-POINT AI GENERATION VERIFICATION CHECKLIST

1. Does the 3D player character render Alexia with white belted tunic, silver ring buckle, long flowing chestnut hair, dark indigo skinny jeans, and cream espadrille flats?
2. Does pressing 'V' cleanly toggle between First-Person and Third-Person view without camera head clipping?
3. Does pressing 'T' toggle the free-flying tactical reconnaissance drone at 80m altitude?
4. Do the zombies exhibit rotting green/purple flesh, exposed ribcage with 3D rib bones, and snarling jaws with fangs?
5. Do the zombies have an asymmetric shambling limp and drag a bare rotting foot across the pavement?
6. Do dead zombies collapse flat into an expanding dark blood pool decal (0x3d0707)?
7. Do all 3 NPC survivors (Dr. Evelyn Reed, Marcus Vance, Sgt. Darius Cole) render with distinct role gear?
8. Does the diesel generator create an interactive expanding base heat dome that pulls distant zombies?
9. Does the Web Audio API synthesizer generate gunfire, heartbeat, generator rumble, and lowpass infection muffling?
10. Does clicking 'RESTART GAME' on the HUD or game-over screen purge all corpses and cleanly restart Day 1?
11. Do the city trees feature a bifurcated forked trunk, grooved bark texture, weeping right branch, and wind-swayed foliage?
12. Do the urban buildings render as 3-story red brick townhouses with concrete plinths, stone storefront bays with pilasters and cornice, front arched windows, side stoop stairs, roof parapets with coping, brick chimneys, and rooftop HVAC units?
13. Does leaving the game screen, pressing ESC, or clicking EXIT GAME trigger an onExit event, display the 'You are out!' popup over a frozen screen, countdown to return to MainMenu, and allow starting a fresh session via 'Start New Game'?